

Dennis Baidoo, MS, GStat

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PROFESSIONAL SUMMARY

Early-career statistician with over four years of experience in data analysis, statistical modeling, and public health research. Skilled in applying advanced statistical techniques to address complex health challenges, with a proven track record of analytical reasoning, adaptability, and effective collaboration across teams. Seeking to leverage leadership, strategic planning, and analytical expertise to advance data-driven solutions in public health and related fields.

EDUCATION

- **University of New Mexico** **Albuquerque, NM**
M.S in Statistics *January 2023 - May 2025*
 - Research: Bayesian Hierarchical Methods for Surveillance of Cervical Dystonia Treatments.
 - Sigma Xi, The Scientific Honor Research Society - Inducted for outstanding achievement in scientific research, reserved for top scholars in the field.
 - Golden Key International Honor Society - Top 15% of the graduating class.
- **University of Ghana** **Accra, Ghana**
B.S in Statistics and Mathematics (Double Major) *Aug 2017 - Nov 2021*
 - Research: Predictors of Intent to Receive COVID-19 Vaccines in Ghana.
 - Dean's Honor List - Top 5% of the graduating class.

PROFESSIONAL EXPERIENCE

- **Algoverse AI Research Program, Algoverse** **Albuquerque, NM**
AI Research Fellow, Advisor: Mr. Kevin Zhu *August 2025 - Current*
 - Selected for a competitive AI research program focused on applying machine learning to real-world problems in healthcare, ethics, and social impact.
 - Collaborated with AI researchers from institutions including Meta, Stanford, OpenAI, Berkeley, and Cornell to develop socially responsible applications of LLMs and AI models.
 - Participated in lectures on algorithmic fairness, bias in medical AI systems, and ethical deployment of large language models in public health contexts.
 - Designed and implemented a research project aligned with public health or health equity, applying AI methods to low-resource settings, bias detection, or medical data.
 - Developed skills in reproducible research, literature review, coding, and presenting findings in alignment with best practices in responsible AI.
 - Earned a merit-based tuition scholarship and support toward potential publication in venues such as the Conference on Neural Information Processing Systems (NeurIPS) or Empirical Methods in Natural Language Processing (EMNLP).
- **College of Population Health, University of New Mexico** **Albuquerque, NM**
Graduate Researcher, Advisor: Dr. Stephanie McIver *May 2024 - Current*
 - Leading mortality burden analysis for student suicide in New Mexico State, using quantitative methods to drive health policy adjustments.
 - Designed predictive machine learning (92% accuracy) and time series models for NIH-funded research study identifying student suicide risk factors, directly informing the university's mental health outreach programs.
 - Identified and ranked critical suicide risk factors (e.g., academic pressure, social isolation), enabling data-driven prevention strategies that reduced high-risk cases.
 - Co-designed suicide prevention talks with campus health teams, embedding mental health modules in statistics courses, and post-intervention surveys showed more than a 30% increase in student help-seeking behavior.

- Submitted findings to a peer-reviewed public health journal, currently under review, to expand the evidence base for university-level suicide prevention initiatives nationwide.

- **Office of the Vice President for Research, University of New Mexico**
Zancada Graduate Fellow, Advisor: Prof. Melissa Emery Thompson

Albuquerque, NM
May 2024 - June 2025

- Entrusted with promoting interdisciplinary research, ethical scholarship, and community engagement to address complex global challenges.
- Participated in bi-weekly professional development workshops focused on leadership, policy engagement, and research communication.
- Collaborated with fellows across diverse disciplines to design actionable research strategies and promote civic impact.
- Strengthened skills in research leadership, academic networking, and translational communication.

- **Center for Teaching and Learning , University of New Mexico**
Graduate Teaching Scholar, Advisor: Dr. Stephanie Spong

Albuquerque, NM
August 2023 - June 2024

- Designed and implemented a Scholarship of Teaching and Learning (SoTL) based classroom intervention; prepared final reflection and presentation for campus-wide showcase.
- Committed to 10+ hours monthly for seminars, consultations, peer observations, and project work on the Scholarship of Teaching and Learning (SoTL).
- Participated in monthly workshops focused on applying evidence-based teaching strategies across disciplines.
- Analyzed mixed-methods data and produced actionable recommendations adopted to enhance TA training and cross-cultural teaching support across U.S. universities.
- Completed peer consultations and teaching observations aimed at improving classroom practices.

- **Department of Anesthesiology, University of New Mexico**
Graduate Research Assistant, Advisor: Dr. Reza Ehsanian

Albuquerque, NM
June 2024 - January 2025

- Supported data collection and analysis for a multi-site study on genicular nerve radiofrequency ablation (RFA) practice patterns, contributing to insights that optimized clinical protocols for 500+ annual procedures.
- Streamlined research collaboration by developing and maintaining centralized project documentation systems, ensuring 100% compliance with institutional protocols and improving team efficiency by 25%.
- Authored strategic briefs synthesizing research outcomes, influencing 3 departmental policy changes in 2024.
- Presented study findings to multidisciplinary clinical teams, fostering cross-specialty dialogue and driving adoption of evidence-based practice changes.

- **African American Student Services, University of New Mexico**
Research and Project Assistant, Advisor: Dr. Brandi Stone

Albuquerque, NM
June 2023 - Current

- Led a high-impact mathematics and statistics pilot program, leveraging data-driven analysis to optimize quantitative assessments, resulting in a 45% increase in student success rates.
- Trained 15+ students in R for qualitative research, boosting reproducibility and enabling 3 conference presentations.
- Boosted research grant success rate from 40% to 80%, while maintaining exceptional program pass rates, driving sustainable funding and impact.
- Developed and implemented targeted mentorship strategies, improving retention rates for underrepresented students by 30%.

- Department of Statistics and Actuarial Science, University of Ghana**
Research Assistant, Advisor: Dr. Godwin Debrah

Accra, Ghana
Oct 2021 - Nov 2022

 - Conducted data analysis for an NIH-funded study on Broken Heart Syndrome, identifying key risk factors that informed clinical screening guidelines.
 - Trained 33+ students in predictive modeling (regression diagnostics, covariate selection), boosting research skills for public health/econometrics projects.
 - Mentored 4 students in statistics/biostatistics final year projects, improving methodological rigor and resulting in 2 publications.
 - Presented analytical findings at departmental seminars, fostering interdisciplinary collaboration and integrating results into ongoing cardiovascular research initiatives.

- Compassion International Ghana**
Intern - Data Analyst, Advisor: Mr. Patrick Attah-Darkoh

Accra, Ghana
May 2018 - Jul 2018

 - Transformed malaria program outcomes through data-driven insights, optimizing interventions that reduced cases by 37% in target regions.
 - Partnered with Compassion International Ghana to optimize resource allocation for malaria-affected children, reducing health disparities in underserved communities by 25%.
 - Designed and implemented dashboards for real-time tracking of key health indicators, enhancing decision-making and stakeholder reporting efficiency.
 - Conducted spatial analysis of malaria incidence to identify high-risk clusters, guiding targeted intervention strategies and improving program impact.

ACADEMIC SERVICE, LEADERSHIP, & VOLUNTEERING

2025 - Reviewer, Journal of Open Source Education (JOSE)
 2025 - Reviewer, Biostats4You: University of Minnesota
 2025 - Reviewer, Undergraduate Statistics Project Competition: The Consortium for the Advancement of Undergraduate Statistics Education (CAUSE) and the American Statistical Association (ASA)
 2023|24 - Reviewer, Ronald E. McNair Scholars Research Competition Conference
 2023|24 - Research Proposal Reviewer, Graduate and Professional Student Association: University of New Mexico
 2023 - Registration Assistant, Biopharmaceutical Section Regulatory-Industry Statistics Workshop: American Statistical Association (ASA)
 2023 - Peer Mentor, Compassion International Ghana
 2023 - Mentor, African American Student Services: University of New Mexico
 2023 - Peer Mentor, Project for New Mexico Graduates of Color: University of New Mexico
 2023 - Student Advisor, Global Education Office: University of New Mexico
 2023 - Organizer, Pentecostal Students and Associates: University of New Mexico
 2023 - Math Competition Coordinator, Statewide Math Contest: University of New Mexico
 2020 - Pandemic Educator, Department of Psychology: University of Ghana

RESEARCH EXPERIENCE

Published Article (Link to Google Scholar)

- Okeke, O. J., **Baidoo, D.**, Frimpong, J.A., et al (2023). Modelling Land Suitability for Optimal Rice Cultivation in Ebonyi State, Nigeria: A Comparative Study of Empirical Bayesian Kriging and Inverse Distance Weighted Geostatistical Models. <https://doi.org/978-624-6045-02-9>

Manuscript Submitted/In-progress

- **Baidoo, D.**, Amevor, R., Kubuafor, E., et al (2025, in review). Bayesian Hierarchical Methods for Surveillance of Cervical Dystonia Treatments. <https://arxiv.org/abs/2508.15762>
- Kubuafor, E., **Baidoo, D.**, Amevor, R., et al (2025, in review). Latent spatial heterogeneity in U.S. cancer mortality: A multi-site clustering and spatial autocorrelation analysis. <https://arxiv.org/abs/2508.12188>
- Amevor, R., **Baidoo, D.**, Kubuafor, E., et al (2025, in review). Integrative Prognostic Modeling of Breast Cancer Survival with Gene Expression and Clinical Data. <https://arxiv.org/abs/2508.16740>

- **Baidoo, D.**, Aboagye, F., Okeke, O. J., et al (2024, in review). Comparing malaria trends in the Comoros Islands: ARIMA modeling and retrospective analysis. <https://arxiv.org/abs/2508.16018>
- **Baidoo, D.**, Okeke, O. J., Amevor, R., et al (2024, in review). Elucidating Community-Level Determinants of Adolescent Obesity in New Mexico: An Artificial Intelligence Approach Using Bayesian Belief Networks.
- Okeke, O. J., **Baidoo, D.**, Okeke, H. U., et al (2024, in review). Modelling the Spatial Distribution of West Nile Virus in New Mexico: A Five-Year Study Employing Poisson, Geometric, Quartile Normalization and Mean Squared Error Evaluation.
- Kubuafor, E., **Baidoo, D.**, Amevor, R., et al (2024, in review). Evaluation of Time Series Forecasting Models for Predicting Lung Cancer Mortality Rates in the United States: A Comparison with Altuhaifa (2023) Study. <https://arxiv.org/abs/2508.16052>
- Okeke, O. J., Enyinnaya, R. N., Nwaogu, C., **Baidoo, D.**, et al (2023, in review). Irrigation Agriculture Potential Areas Extraction in Hadejia Jama'are River Basin Using Multivariate Clustering Geostatistical Analysis.
- Okeke, O. J., Nwaogu, C., Oluwadamilare, O. O., **Baidoo, D.**, et al (2023, in review). Modelling the Groundwater Potential and Suitability for Irrigation in New Mexico State Using Geospatial, MLA and Geostatistical Approaches.

Poster Presentation

- **Baidoo, D.** (2025, October). Bayesian Hierarchical Methods for Surveillance of Cervical Dystonia Treatments. Paper accepted for presentation at the International Forum on Research Excellence, Sigma Xi.
- **Baidoo, D.** (2025, August). Bayesian Hierarchical Methods for Surveillance of Cervical Dystonia Treatments. Paper accepted for presentation at the Statistics in Pharmaceuticals (SIP), University of Connecticut.
- **Baidoo, D.**, & Okeke, O. J. (2024, February). Leveraging Bayesian Belief Networks to Uncover Place-Based Influences on Adolescent Obesity in New Mexico. Paper accepted for presentation at the American Statistical Association Conference on Statistical Practice, New Orleans, Louisiana.
- **Baidoo, D.**, & Okeke, O. J. (2023, September). Leveraging Bayesian Belief Networks to Uncover Place-Based Influences on Adolescent Obesity in New Mexico. The paper was accepted for presentation at the Prof. Michael Woodroffe Memorial Conference, Ann Arbor, Department of Statistics, University of Michigan.
- **Baidoo, D.**, & Okeke, O. J. (2023, November). Modelling The Groundwater Potential and Suitability For Irrigation In New Mexico State Using Geospatial MLA and Geostatistical Approaches. The paper was accepted for presentation at the GIS Day, Albuquerque, University of New Mexico.

Talks

- **Baidoo, D.** Fundamentals of Quantitative Research. University of Ghana, Department of Statistics and Actuarial Science. Accra, February 2025.
- **Baidoo, D.** Crossing Academic Borders: Navigating Teaching in the U.S. as an International Student. Center for Teaching and Learning: University of New Mexico, Albuquerque, May 2024.
- **Baidoo, D.** Intro to R. University of Ghana, Department of Statistics and Actuarial Science. Accra, June 2024.
- **Baidoo, D.** Intro to Latex. University of Ghana, Department of Statistics and Actuarial Science. Accra, October 2023.

TEACHING EXPERIENCE

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| <ul style="list-style-type: none"> • Department of Mathematics and Statistics, University of New Mexico
 Course Instructor <ul style="list-style-type: none"> ◦ Designed and taught the first-ever offering of the new MATH 1300 (Statistical Literacy) course from scratch to non-quantitative majors. ◦ Designed and delivered 350+ engaging lectures for MATH1350 (Introductory Statistics), reaching 500+ undergraduates across 7 semesters - Spring 2023, Fall 2023, Spring 2024, Summer 2024, Spring 2025, Summer 2025, and Fall 2025. | Albuquerque, NM
Jan 2023 - Current |
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- Cultivated a welcoming classroom atmosphere that encouraged questions and collaborative learning.
- Guided 20+ final-year students in applying descriptive and inferential statistics to psychology research projects.
- Reduced departmental dropout rates by 66% through proactive student support interventions.

- **Department of Statistics and Actuarial Science, University of Ghana** **Accra, Ghana**
Teaching Assistant, Instructor: Dr. Godwin Debrah **Oct 2021 - Nov 2022**
 - Led rigorous yet accessible instruction in STAT 445 (Regression Analysis - linear, logistic, nonlinear) and time series analysis for mixed audiences of quantitative and non-quantitative majors.
 - Provided 10+ hours weekly of one-on-one statistical guidance, helping 86% of students successfully implement regression analysis in their capstone projects.
 - Maintained a high passing rate with more than 90% getting an A with no students failing the course.
 - Integrated real-world datasets and applied case studies into lectures, increasing student engagement and practical problem-solving skills.
- **Eastern New Mexico University, Department of Nursing** **Albuquerque, NM**
Course Instructor **May 2023 - July 2023**
 - Taught STAT 1350: Introductory Statistics, delivering lectures and labs on probability, hypothesis testing, regression, and statistical literacy.
 - Created and assessed assignments, quizzes, and exams to evaluate mastery of core statistical concepts.
 - Integrated real-world data sets to illustrate applications in health, business, and social sciences.
 - Implemented interactive teaching strategies (group problem-solving, technology tools) to foster engagement.
- **Mozano Senior High School** **Gomoa Eshiem, Ghana**
Course Instructor **Jul 2019 - Aug 2019**
 - Tutored core and elective mathematics at my alma mater, supporting students in mastering fundamental and advanced concepts.
 - Mentored underrepresented student groups on advancing their studies in mathematical sciences at the university.
 - Mentored 50+ minority students in statistical methods, with 20% pursuing bachelor's degrees in statistics/economics at top universities.
 - Organized peer-led study groups and academic workshops, fostering collaborative learning and improving overall student performance in quantitative courses.

AWARDS & HONORS

2025 - Norman Beery Memorial Scholarship, American Statistical Association
 2025 - Lester R. Curtin Award, American Statistical Association
 2025 - Statistics in Pharmaceuticals (SIP) Scholarship Award, University of Connecticut
 2025 - JSM Student and Early Career Travel Award, American Statistical Association
 2025 - SDSS Student and Early Career Travel Award, American Statistical Association
 2025 - Excellence in Teaching Award, University of New Mexico
 2025 - The South Central Topology Conference IV Travel Award, Texas A & M University
 2024 - Biopharmaceutical Section Student Scholarship Award, American Statistical Association
 2024 - Diversity Mentoring Program Award, Joint Statistical Meetings
 2024 - Office of the Vice President for Research Zancada Graduate Fellowship, University of New Mexico
 2024 - Axiom Diversity Scholarship Award, Axiom
 2024 - Graduate Scholarship Fund Grant, University of New Mexico
 2024 - Graduate Community Mentoring Scholarship, University of New Mexico
 2023|25 - Professional Development Grant, University of New Mexico
 2023|25 - Project for New Mexico Graduates of Color Scholarship, University of New Mexico
 2023 - Scholarship of Teaching & Learning Award, University of New Mexico
 2023 - The South Central Topology Conference III Travel Award, New Mexico State University
 2023 - Prof. Michael Woodroffe Memorial Conference Travel Award, University of Michigan
 2023 - Student Research Grant, University of New Mexico
 2012 - Overall Best Student in HIV/AIDS Prevention Competition, Gomoa West Education District

NON-DEGREE EDUCATION & CERTIFICATES

2025 - University of Connecticut: AI for Clinical Trials and Humanized AI for Future Healthcare
2025 - Joint Statistical Meetings: Cultivating a Career as a Statistical Collaborator and Medical Researcher in the Pharmaceutical Industry
2025 - Joint Statistical Meetings: Statistical Issues with Large Language Models
2025 - Joint Statistical Meetings: Bayesian Precision Medicine
2025 - Joint Statistical Meetings: Bayesian Time Series Analysis and Forecasting
2024 - American Statistical Association: Graduate Statistician accreditation (GStat)
2024 - Johns Hopkins University (Coursera): Clinical Trials Analysis, Monitoring, and Presentation
2024 - Johns Hopkins University (Coursera): Infectious Disease Modeling in Practice
2024 - Johns Hopkins University (Coursera): Practical Machine Learning
2023 - Ronald E. McNair Scholars Research Competition Conference: Outstanding commitment to nurturing undergraduate student growth in research

HIGHLIGHTED SKILLS

Soft Skills: Effective Communication, Team Management, Problem-Solving and Critical Thinking, Interpersonal Skills, and Relationship-Building

Technical Skills: Proficiency in data analysis tools (R, SAS, Python, OpenBUGS), LaTeX, Microsoft 365, Prompt expert with LLMs, Covidence

PROFESSIONAL MEMBER ASSOCIATION

American Statistical Association
Sigma Xi
Golden Key International Honour Society
Eastern North American Region of the International Biometric Society
American Mathematical Society